

Mathematics 629 — Homework 7

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Problem 1

Let $X_1 = [0, 1]$, with $\mathcal{M}_1 = \mathcal{B}_{[0,1]}$ and with μ_1 the Lebesgue measure, $X_2 = [0, 1]$ with $\mathcal{M}_2 = \mathcal{B}_{[0,1]}$ and μ_2 being counting measure. Let $\mu = \mu_1 \times \mu_2$. Consider the diagonal $D = \{(x, x) : 0 \leq x \leq 1\}$.

- (i) Use the definition of μ to evaluate $\int \mathbf{1}_D d\mu$.

Answer. By definition of product measure, $\int \mathbf{1}_D d\mu = \int \mu_2(sl_{x_1}) d\mu_1(x_1)$ where $sl_{x_1} = x_1$. Then $\int \mathbf{1}_D d\mu = \int_{[0,1]} 1 d\mu_1 = 1$. $\square$

- (ii) Show that $\int \int \mathbf{1}_D d\mu_1 d\mu_2$ and $\int \int \mathbf{1}_D d\mu_2 d\mu_1$ have different values. Why can Tonelli's theorem not be applied?

Answer. In (i), integrating with respect to μ_2 first showed $\int \int \mathbf{1}_D d\mu_2 d\mu_1 = 1$. Let $sl_{x_2} = x_2$. Then similarly,

$$\int \int \mathbf{1}_D d\mu_1 d\mu_2 = \int \mu_1(sl_{x_2}) d\mu_2(x_2) = 0 * (sl_{x_2}) = 0.$$

Clearly order of integration matters in this case. Since Tonelli guarantees equivalence regardless of order of integration, the theorem cannot be applied here because $(X_2, \mathcal{M}_2, \mu_2)$ is not sigma finite ($\mu_2([0, 1])$ is uncountable). $\square$

Problem 2

Consider, for $x \in (0, 1) \times (0, 1)$,

$$f(x_1, x_2) = \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} = \frac{\partial^2}{\partial x_1 \partial x_2} \arctan \frac{x_1}{x_2}.$$

- (i) Verify that

$$\int_0^1 \int_0^1 \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} dx_2 dx_1 = \frac{\pi}{4}, \quad \int_0^1 \int_0^1 \frac{x_1^2 - x_2^2}{(x_1^2 + x_2^2)^2} dx_1 dx_2 = -\frac{\pi}{4}.$$

Answer.

□

(ii) Let $T = \{x : 0 < x_2 < x_1 < 1\}$, let m be Lebesgue measure in $(0, 1)^2$ and show that

$$\int_T f \, dm = \infty.$$

Answer.

□

(iii) Why can Fubini's theorem not be applied in (i)?

Answer.

□

Problem 3

Let $(X_1, \mathcal{M}_1, \mu_1)$ be a measure space and $X_2 = \mathbb{N}$ with $\mathcal{M}_2 = \mathcal{P}(\mathbb{N})$ and counting measure μ_2 . Interpret Tonelli's and Fubini's theorem in this situation and show that one does not need the assumption of σ -finiteness of μ_1 in this case.

Answer.

□

Problem 4

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. Let f be a nonnegative measurable function on X and

$$U_f = \{(x, y) \in X \times [0, \infty] : 0 \leq y \leq f(x)\}.$$

Show that U_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$ measurable and $\mu \times m(U_f) = \int f \, d\mu$. Show that the Lebesgue measure (in $\mathbb{R}^2$) of the graph of f (i.e., $\{(x, f(x)) : x \in X\}$) is zero.

Answer.

□

Problem 5

Let

$$A_1 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + z^2 \leq 1\}, \quad A_2 := \{(x, y, z) \in \mathbb{R}^3 : y^2 + z^2 \leq 1\}$$

and compute the Lebesgue measure of $A_1 \cap A_2$.

Answer.

□